

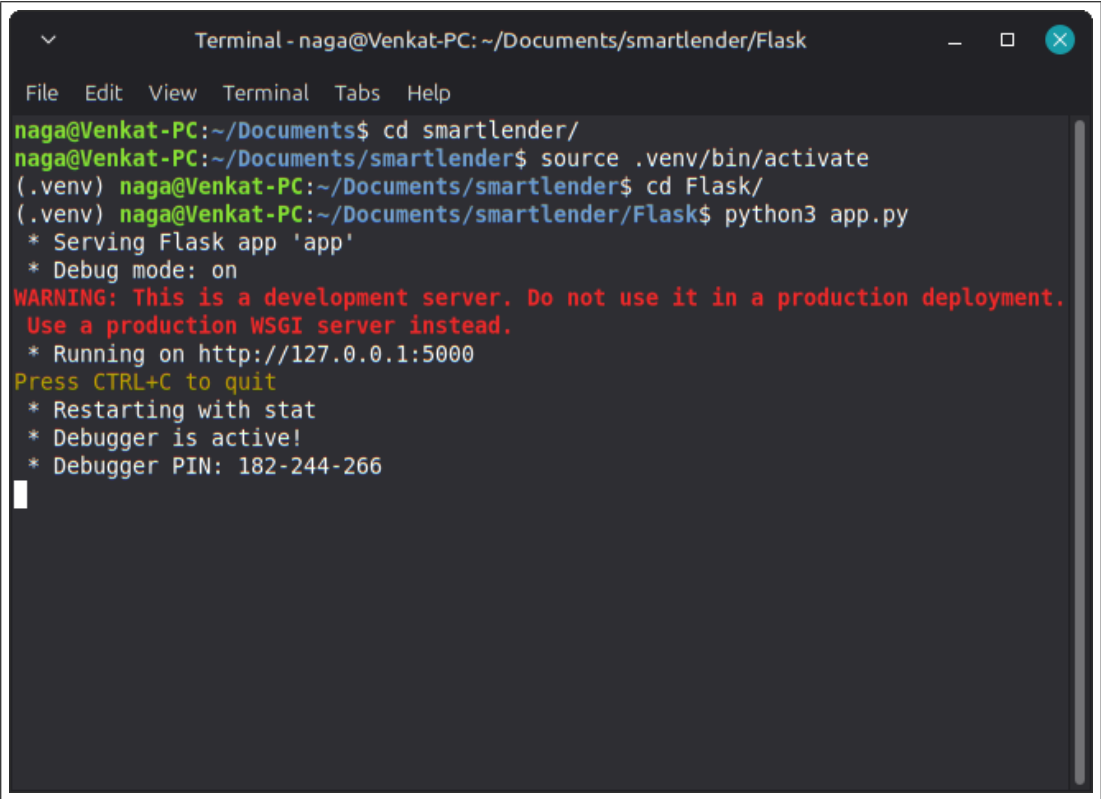
Phase 8: Project Demonstration

Execution & UI Evidence

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Terminal Execution

The following screenshot demonstrates the successful initialization of the Flask development server, indicating the application is ready to accept HTTP requests on local port 5000.

A terminal window titled "Terminal - naga@Venkat-PC: ~/Documents/smartlender/Flask" showing the execution of commands to start a Flask application. The commands include navigating to the smartlender directory, activating the virtual environment, navigating to the Flask subdirectory, and running the app.py file. The output shows the server starting on http://127.0.0.1:5000 with debug mode on. A warning message is displayed: "WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead." The terminal also shows the server restarting with stat, the debugger is active, and the debugger PIN is 182-244-266.

```
Terminal - naga@Venkat-PC: ~/Documents/smartlender/Flask
File Edit View Terminal Tabs Help
naga@Venkat-PC:~/Documents$ cd smartlender/
naga@Venkat-PC:~/Documents/smartlender$ source .venv/bin/activate
(.venv) naga@Venkat-PC:~/Documents/smartlender$ cd Flask/
(.venv) naga@Venkat-PC:~/Documents/smartlender/Flask$ python3 app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 182-244-266
```

Figure 1: Flask Server Running in Terminal

Web Interface - Home Page

This screenshot displays the landing page of the SmartLender application. It outlines the project details, the technology stack utilized, and features the call-to-action button to navigate to the application form.

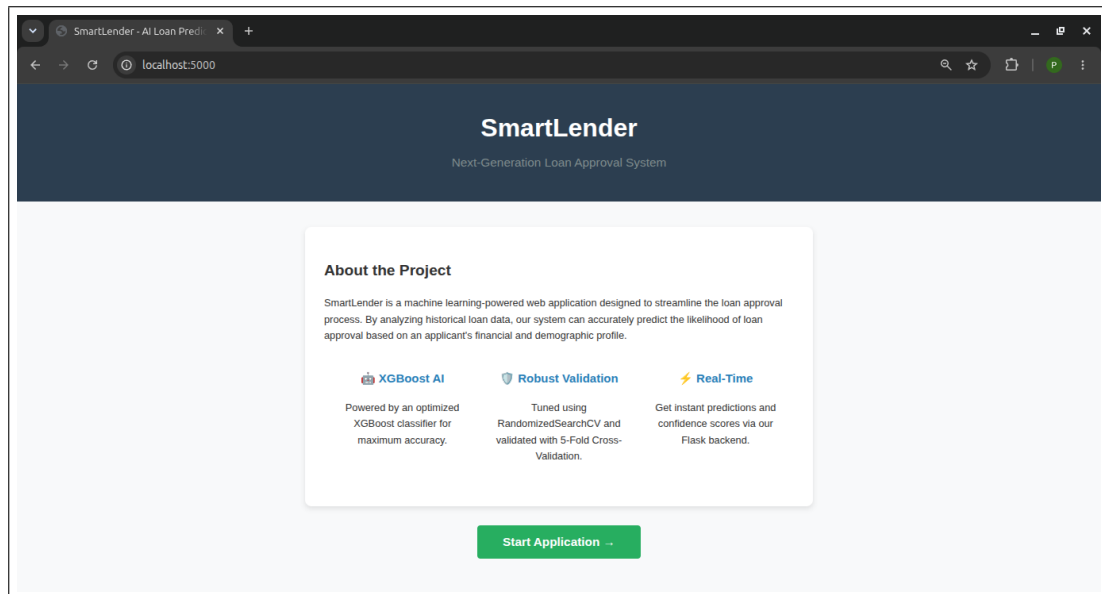
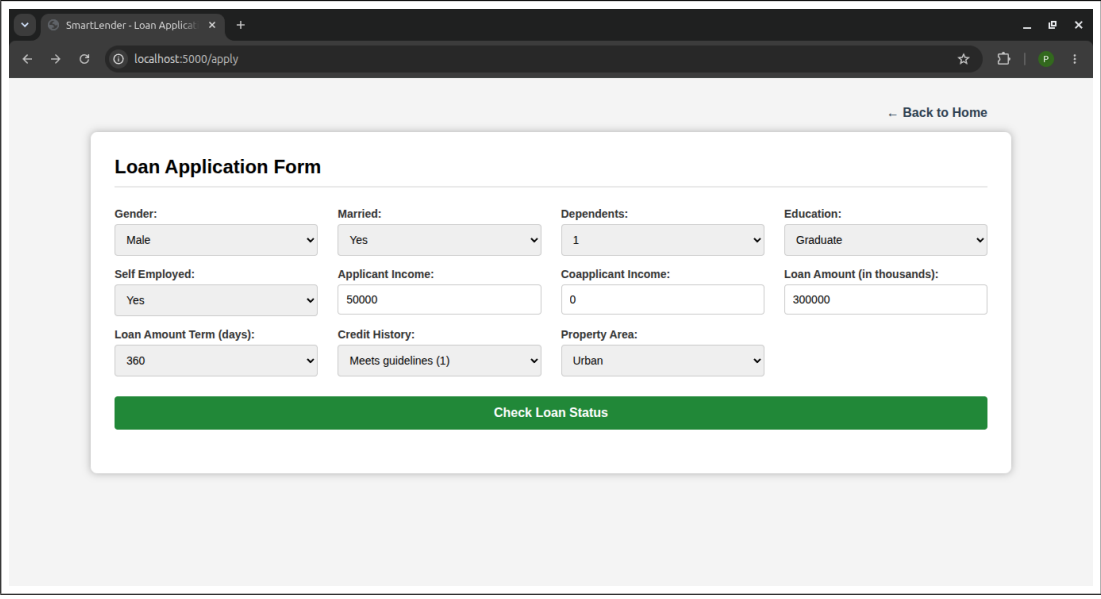


Figure 2: Landing / Home Page Interface

Web Interface - Form Input

This screenshot displays the user filling out the Loan Application Form with various demographic and financial parameters mapped to their respective integer encodings.



The screenshot shows a web browser window with the address bar displaying "localhost:5000/apply". The page title is "SmartLender - Loan Application". A link labeled "-- Back to Home" is located in the top right corner. The main content is a "Loan Application Form" with the following fields:

Gender:	Married:	Dependents:	Education:
Male	Yes	1	Graduate

Self Employed:	Applicant Income:	Coapplicant Income:	Loan Amount (in thousands):
Yes	50000	0	300000

Loan Amount Term (days):	Credit History:	Property Area:
360	Meets guidelines (1)	Urban

At the bottom of the form is a green button labeled "Check Loan Status".

Figure 3: Input Form Interface

Web Interface - Output Result

This screenshot exhibits the system successfully processing the input and rendering the prediction result (APPROVED/REJECTED) along with the XGBoost model's confidence percentage.

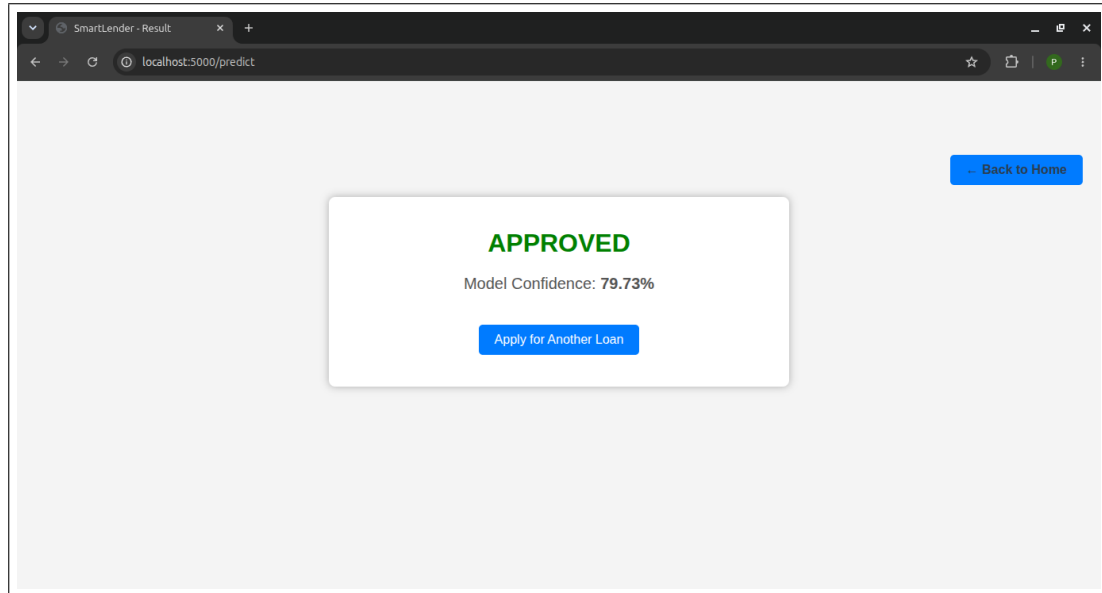


Figure 4: Prediction Output Display

Demo Video Link

For a comprehensive, real-time walkthrough of the system's functionality—from loading the server to submitting the form and viewing results—please refer to the following screen recording:

Video Link: [Insert Google Drive / YouTube Link Here]